

Paper Manager pack — $\pi_{0.5}$ (Sep 30)

Paper: $\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization

arXiv: <https://arxiv.org/abs/2504.16054>

Blog: <https://www.physicalintelligence.company/blog/pi05>

PDF: /workspace/papers/pi05-2504.16054.pdf

Session: Sep 30 · due 9:30 AM · **You = Visionary (seat 1)**

Note: Other role seats empty on schedule — full five packs still included for club/GitHub

Nearby: Oct 2 summary proposal deadline

Sources: Summarizer + Senior Research (/workspace/pi05-quiet/DELIVERABLE.md)

1. Paper summary

Follow-on to π_0 for **open-world** (new homes) via heterogeneous **co-training** + **hierarchical** inference (semantic subtask → continuous FM chunk). Two-stage: **FAST discrete pretrain** (280k) → **FM action expert post-train** (80k). Same model: AR subtask then **10 FM steps**, ~50-step/~1s chunks @ **50 Hz**. ~400h mobile-manip / ~100 homes; claims long-horizon cleaning in entirely new homes.

FM vs FAST myth: Hybrid FAST pretrain + **FM retained** for continuous control. Blogs saying “FAST replaced FM” are **FALSE**. OpenPI ships **FM head** for $\pi_{0.5}$; KI (2505.23705) formalizes stop-grad FAST+FM.

2. Keywords

$\pi_{0.5}$, open-world VLA, co-training (MM/ME/CE/HL/WD/VI), hierarchical subtasks, FAST+FM hybrid, knowledge insulation, flow matching, BEHAVIOR baseline, RTC

3. FM vs FAST flag (say this aloud)

Claim	Verdict
FAST replaced FM	**MYTH**
FAST pretrain + FM post-train	**TRUE**
FM head retained at infer	**TRUE**
OpenPI FM API for $\pi_{0.5}$	**CLEAR**
FAST is the deployed low-level controller	**FALSE**

4. Role packs

Stakeholder (18+3)

Open-world KPI = **new homes**, not one kitchen. OpenPI $\pi_{05_}$ * + BEHAVIOR'26 official baseline (with GR00T N1.7). BEHAVIOR'25 winner on Pi0.5 ($q \approx 0.260$) validates substrate. Budget FM fine-tunes + optional RTC; ignore FAST-replaced-FM myths.

Scientific Reviewer (8+3)

Strong system paper; unseen protocol + ablations. Caveats: figure-only bars; abbreviated scaling recipe; private MM data; human-HL oracle result surprising; KI clearer in follow-up. Hybrid FAST → FM must be cited accurately.

Empiricist (11+3)

- H1: $\pi_{0.5} > \pi_0$ on multi-stage/lang under matched FT
- H2: HL on > flat prompt → action on long chains
- H3: WD/semantic co-train helps OOD lang; robot diversity helps OOD scenes
- H5: mid horizon ~32 often beats raw H=50 open-loop on BEHAVIOR-like tasks

Priority	Experiment
P0	π_0 vs $\pi_{0.5}$ (public OpenPI/Comet); HL on/off; held-out scene proxy
P1	Co-train ablate (WD/ME/CE); location scaling; $H \in \{8,16,32,50\}$
P2	NFE grid; VI/HL quality proxy
Defer	Full private MM recipe

Budget: eval 12–24 GB; short FT 24–48 GB. Env: /workspace/pi05-quiet/env_outline/.

Archaeologist (11+3)

RT-1/2, OpenVLA, Octo, OXE → π_o (FM) → **FAST** → $\pi_{o.5}$ (open-world+hybrid+HL) → **KI** → **RTC** → Hi Robot. BEHAVIOR-1K → '25 (Pi0.5 winner) → '26 ($\pi_{0.5}$ baseline). What's new: mixture, unified HL/LL, VI, home-scale eval.

Visionary (11+3) — ****YOU (seat 1)**** — HIGH PRIORITY

2026 BEHAVIOR (CLEAR)

- Baselines: **$\pi_{0.5}$ + GR00T N1.7**
- 100 tasks / 7 scenes / 20k demos
- Launch ~07/02/2026 · submit **10/16/2026** · winners ~11/04/2026
- '25 winner adapted Pi0.5 (correlated FM noise, System-2 stages, multi-sample FM, compression) → ~26% q — preview of '26 bottlenecks (grasps, non-Markov stages, recovery)

Oct 2 proposal framing: $\pi_{0.5}$ -baseline + targeted deltas, not greenfield.

High-leverage deltas to champion

1. **Keep FM**; FAST/KI for reps/language — never “replace FM with FAST”
2. Hierarchy / stage tracking (paper HL + '25 System-2)
3. **RTC** for delayed control loops
4. Recovery / VI-style intervention data
5. Cross-embodiment + web/HL co-training when scene diversity is limited

Apps: new-home cleaning, language tidying, coachable robots, OpenPI FT on DROID/LIBERO/custom.

Caution: real-home success ≠ solved BEHAVIOR (BDDL, depth, 100-task multitask). Still clearest shared substrate linking PI open-world, OpenPI, and 2026 baseline.

Skip Discord/Drive.

5. Control note

Hierarchical subtask tokens = semantic outer loop; FM action chunks = inner control. Open-world = held-out scene generalization via co-train mix — **not** a classical planner. RTC is the companion paper for async soft-inpaint.

6. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Open-world KPI + OpenPI/BEHAVIOR'26
Reviewer	8+3	Strengths + private-data / citation hygiene
Empiricist	11+3	π_0 vs $\pi_{0.5}$ + HL + horizon
Archaeologist	11+3	$\pi_0 \rightarrow \text{FAST} \rightarrow \pi_{0.5} \rightarrow \text{KI} \rightarrow \text{RTC} \rightarrow \text{BEHAVIOR}$
Visionary	**11+3**	**2026 map + Oct 2 proposal deltas (HIGH)**

Spine: open-world + hybrid FAST → **FM retained**. PDF for Mac. All five roles for GitHub.

Paths

Artifact	Path
This pack	`/workspace/pi05-quiet/KANTA_PACK.md`
Senior Research	`/workspace/pi05-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/pi05-2504.16054-summary.md`
Paper PDF	`/workspace/papers/pi05-2504.16054.pdf`